

Mathematical Derivations

Full derivations of the four relations stated in Section V of the manuscript. All four are corollaries of Parseval's identity for a unitary transform. They are elementary; they are set out in full because their *interpretive* consequences drive the paper's conclusions.

D1. Preliminaries

Let $x[n]$ be real, $n = 0, \dots, N-1$, with sample mean $\mu = (1/N)\sum x[n]$ and population variance $\sigma^2 = (1/N)\sum (x[n]-\mu)^2$. Under the unitary convention

$$X[k] = (1/\sqrt{N}) \sum_n x[n] \exp(-j2\pi kn/N),$$

the transform matrix is unitary, hence $E = \sum_n |x[n]|^2 = \sum_k |X[k]|^2$.

D2. The DC energy share (Equation 4)

Setting $k = 0$ in the analysis equation gives $X[0] = (1/\sqrt{N})\sum_n x[n] = \sqrt{N} \mu$, so $|X[0]|^2 = N\mu^2$.

For the total energy, expand about the mean:

$$E = \sum_n x[n]^2 = \sum_n ((x[n]-\mu) + \mu)^2 = \sum_n (x[n]-\mu)^2 + 2\mu \sum_n (x[n]-\mu) + N\mu^2.$$

The cross term vanishes because $\sum (x[n]-\mu) = 0$ by definition of the mean, leaving $E = N\sigma^2 + N\mu^2 = N(\mu^2 + \sigma^2)$. Therefore

$$\eta_{DC} = |X[0]|^2/E = N\mu^2 / N(\mu^2 + \sigma^2) = \mu^2/(\mu^2 + \sigma^2).$$

Dividing numerator and denominator by σ^2 and writing $r = \mu/\sigma$ gives $\eta_{DC} = r^2/(1+r^2)$. $\square$

The function is monotone increasing in $|r|$ with $\eta_{DC}(0) = 0$, $\eta_{DC}(1) = 1/2$, $\eta_{DC}(3) = 0.9$ and $\eta_{DC}(10) = 0.990$. It also shows that the AC share is $1/(1+r^2)$, which decays as r^{-2} . For the recovered baseline, $r = 3.6973$ gives $\eta_{DC} = 0.9318337$, matching the measured value to 3×10^{-16} .

D3. Error from retention (Equation 7)

Let K be a retention set and $\hat{x}$ the inverse transform of the masked spectrum. By linearity of the inverse transform and unitarity,

$$\|x - \hat{x}\|^2 = \|X - \hat{X}\|^2 = \sum_{k \notin K} |X[k]|^2 = E - \sum_{k \in K} |X[k]|^2 = (1-\eta)E,$$

where the second equality holds because the retained and discarded index sets are disjoint, so the

corresponding subspaces are orthogonal. Dividing by $\|x\|^2 = E$ and taking the square root:

$$\varepsilon_{L2} = \sqrt{(1 - \eta)}. \quad \square$$

Interpretive consequence. The relation holds for *any* orthonormal transform and for η and ε computed on the same basis. It follows that retained energy and relative L2 error are the same information under a monotone transformation. Reporting both therefore cannot corroborate either, and the remedy of "quoting a reconstruction error alongside the energy figure" is ineffective. Verified against 324 independent retention measurements; maximum deviation 1.0×10^{-14} .

D4. Basis conversion (Equation 8)

Suppose the DC bin is retained, $K \ni 0$. Write $E_{DC} = |X[0]|^2$ and $E_{AC} = E - E_{DC}$. The retained energy splits as

$$\sum_{k \in K} |X[k]|^2 = E_{DC} + \sum_{k \in K, k \neq 0} |X[k]|^2 = E_{DC} + \eta_{AC} E_{AC},$$

by definition of η_{AC} . Hence

$$1 - \eta_{total} = (E - E_{DC} - \eta_{AC} E_{AC})/E = E_{AC}(1 - \eta_{AC})/E = (1 - \eta_{AC})(1 - \eta_{DC}),$$

and substituting D2 gives

$$1 - \eta_{total} = (1 - \eta_{AC})/(1 + r^2). \quad \square$$

The two reporting bases are therefore related by a factor determined entirely by r . This is why the reporting protocol requires r to be stated: it is a *sufficient statistic* for converting a published retention figure into the other convention. Maximum deviation over the corpus and offset sweep: 2.2×10^{-16} .

D5. Error scaling (Equation 9)

Apply D3 on each basis: $\varepsilon_{full} = \sqrt{(1 - \eta_{total})}$ and $\varepsilon_{AC} = \sqrt{(1 - \eta_{AC})}$. Substituting D4,

$$\varepsilon_{full} = \sqrt{((1 - \eta_{AC})/(1 + r^2))} = \varepsilon_{AC}/\sqrt{(1 + r^2)}. \quad \square$$

For the recovered baseline $\sqrt{(1 + r^2)} = 3.8301$: an error of 34.80% on the fluctuation is reported as 9.08% on the full signal. Maximum deviation: 1.1×10^{-16} .

D6. Why the modulation operation is not an inverse

The earlier work labels $x1[n] = (-1)^n x[n]$ as "the inverse of the function". Since $(-1)^n = \exp(j\pi n) = \exp(j2\pi(N/2)n/N)$ for even N , this is modulation by the Nyquist frequency. By the modulation theorem,

$$X_1[k] = (1/\sqrt{N}) \sum_n x[n] \exp(j2\pi(N/2)n/N) \exp(-j2\pi kn/N) = X[(k - N/2) \bmod N],$$

a circular shift of the spectrum by $N/2 = 13$ bins. This was confirmed numerically: the coefficient table of the earlier work is its own earlier table rolled by exactly 13 positions. The operation is correct and instructive; it is not an inverse of anything, and the DFT, the inverse DFT, modulation, spectral shift and reconstruction are distinguished strictly throughout the manuscript.

D7. On "verifying" Parseval's identity

Under a unitary transform, $E = H$ is an algebraic consequence of unitarity and holds for every input. A numerical procedure that computes both sides and compares them therefore *cannot produce a negative result*, and has no empirical content: it discriminates between no hypotheses. What such a computation does measure is the accumulation of floating-point error, which is a property of the arithmetic and the implementation rather than of the theorem. That quantity is reported in Section VIII-J as ε_p , and its scale relative to the modeling error is the substantive finding.